BT

BREATHTWIN

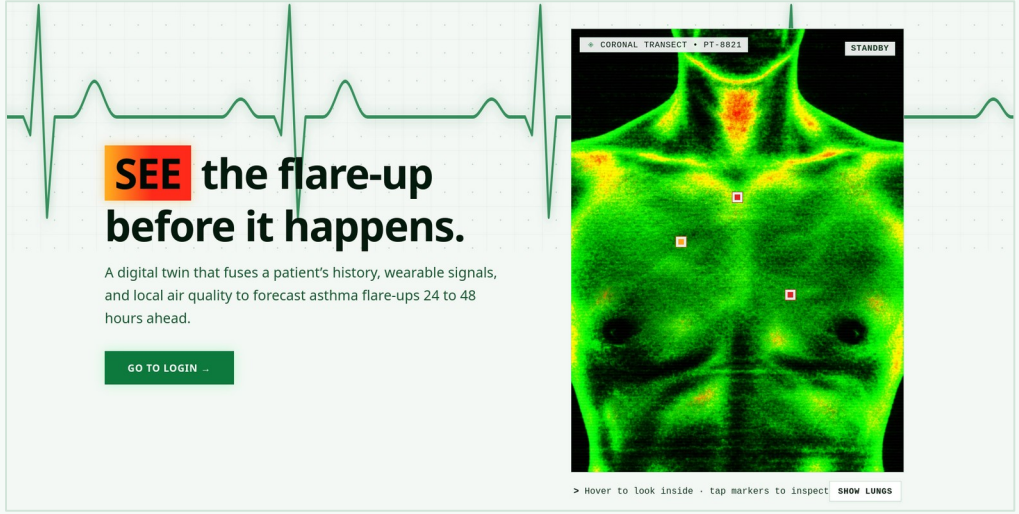
MULTIMODAL ASTHMA DIGITAL TWIN

PROJECT PRESENTATION

BreathTwin

A multimodal digital twin for forecasting asthma flare-up risk 24–48 hours ahead

Connect clinical history, wearable signals and outdoor air quality. Estimate a hidden airway state, forecast risk, and let clinicians explore “what-if” scenarios.



DOCTOR + PATIENT PORTALS

LNMIIT × NSUT

Aarush Bhattacharjee · **Team lead & full-stack / backend**

Rakshit Gupta · **Machine learning**

Open source • MIT License • Synthetic patient cohort

PROJECT REPOSITORY · github.com/Frisbee148/DigitalTwin

THE NEED

The warning signals are spread across different places

A clinician may see the pieces at different times: a baseline in the record, a change in wearable readings, and a change in outdoor air quality.

01

Clinical history

What is normal for this patient?

- Spirometry and lung function
- Controller regimen and adherence
- Previous flare-ups and care history

02

Wearable signals

What is changing right now?

- Overnight oxygen and heart rate
- Resting respiratory rate and HRV
- Sleep, symptoms and rescue use

03

Outdoor conditions

What is the patient exposed to?

- PM2.5 and PM10 concentrations
- Air Quality Index (AQI)
- Temperature and humidity

BreathTwin's goal Bring these signals together and give the care team time to review a risk estimate before an emergency becomes the first clear signal.

THE APPROACH

One connected workflow, from signals to a useful next step

BreathTwin combines learned predictions with a mechanistic airway simulation, then presents the result through doctor and patient interfaces.

1

COLLECT

Clinical baseline, 20-channel wearable window, and local air-quality data.

2

ESTIMATE

A modelled airway-inflammation state that changes with exposure and controller adherence.

3

FORECAST

24-hour and 48-hour flare-up risk, plus a severity estimate.

4

EXPLORE

Compare simulated outcomes under different prevention scenarios.

5

COMMUNICATE

Doctor triage, explainable drivers, and plain-language patient guidance.

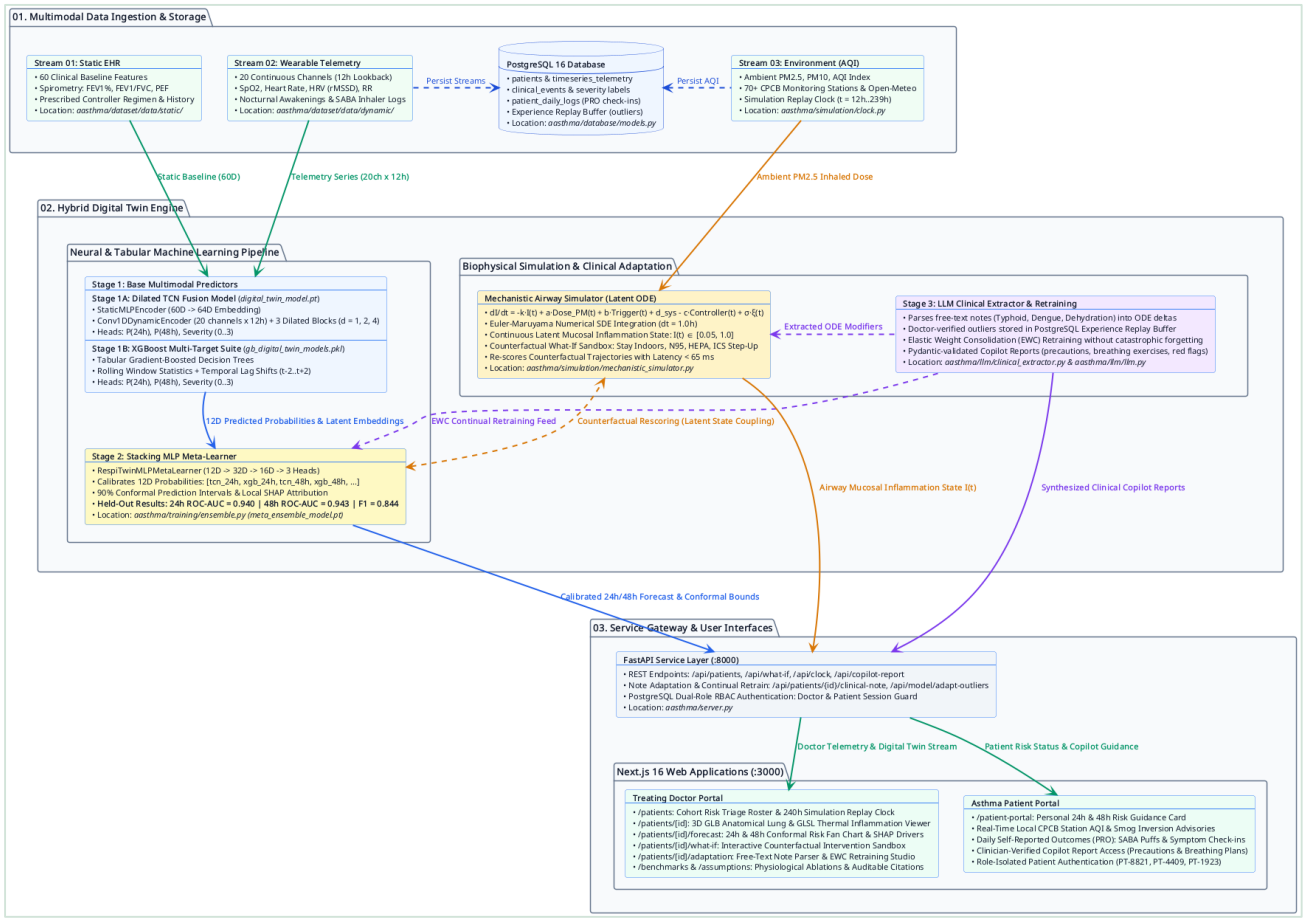
What makes it a digital twin? It is not only a risk score. The system maintains a modelled airway state, updates it as inputs change, and re-runs the forecast when a clinician tests a scenario.

Clinical decision support only — not a diagnostic or prescribing system.

SYSTEM ARCHITECTURE

Three layers keep the project understandable

The diagram shows how incoming data feeds the hybrid twin engine, then reaches the application interfaces through the API layer.



01 **Input & storage**

Static health records, wearable telemetry, environmental readings and PostgreSQL persistence.

02 **Hybrid twin engine**

Dilated TCN + XGBoost + stacking MLP, supported by a mechanistic airway ODE and note adaptation.

03 **Services & interfaces**

FastAPI endpoints serve doctor triage, forecasts, what-if tools and the patient companion.

Architecture image supplied with the project

DATA INGESTION

Three streams describe the patient and their environment

Inputs are aligned into a common view; the clinical history and hourly telemetry provide the patient context while air quality describes an external exposure.

STREAM 01

Static clinical baseline

60 features

Patient history and baseline status

- Age, sex, BMI and smoking history
- FEV1, FEV1/FVC and peak flow
- Controller regimen and adherence
- Past flare-ups and urgent care

STREAM 02

Wearable telemetry

20 channels × 12 h

A rolling window of recent signals

- SpO2, heart rate, respiratory rate, HRV
- Skin temperature and activity
- Sleep and nocturnal awakenings
- Symptoms and rescue inhaler use

STREAM 03

Environmental context

PM2.5 / PM10 / AQI

Local air and weather conditions

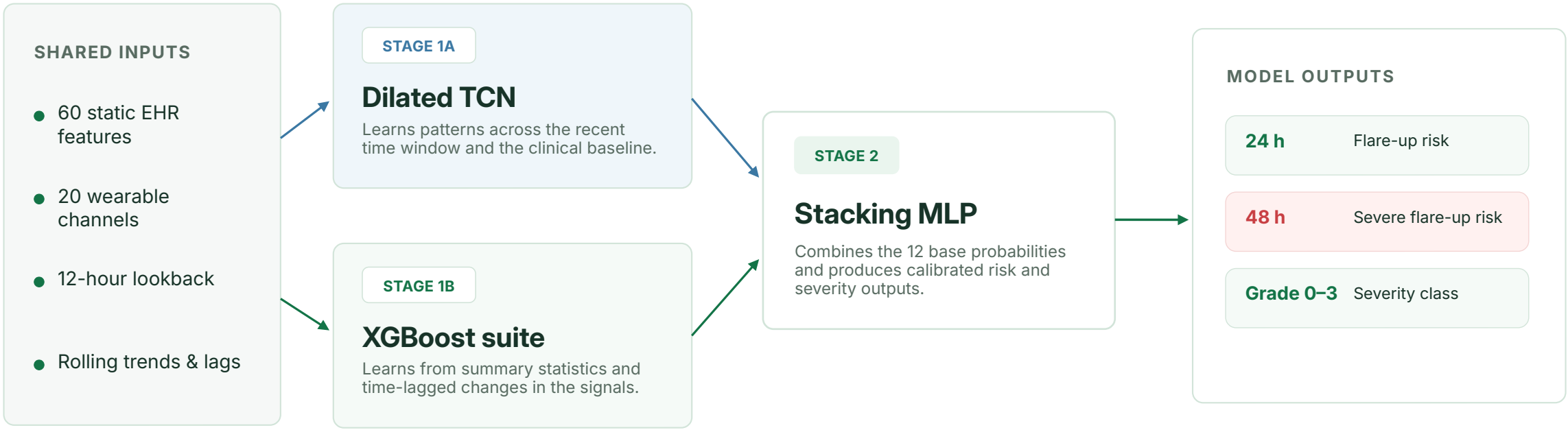
- Particulate matter levels
- AQI from monitoring stations
- Temperature and humidity
- Live feed or replayed conditions

POSTGRESQL 16 Stores patient records, time-series telemetry, clinical events, daily logs and user accounts.

PREDICTIVE PIPELINE

Two base models learn different patterns; a third combines their outputs

The ensemble uses recent time-series behaviour and engineered tabular features, then stacks their probabilities into a single multi-horizon estimate.

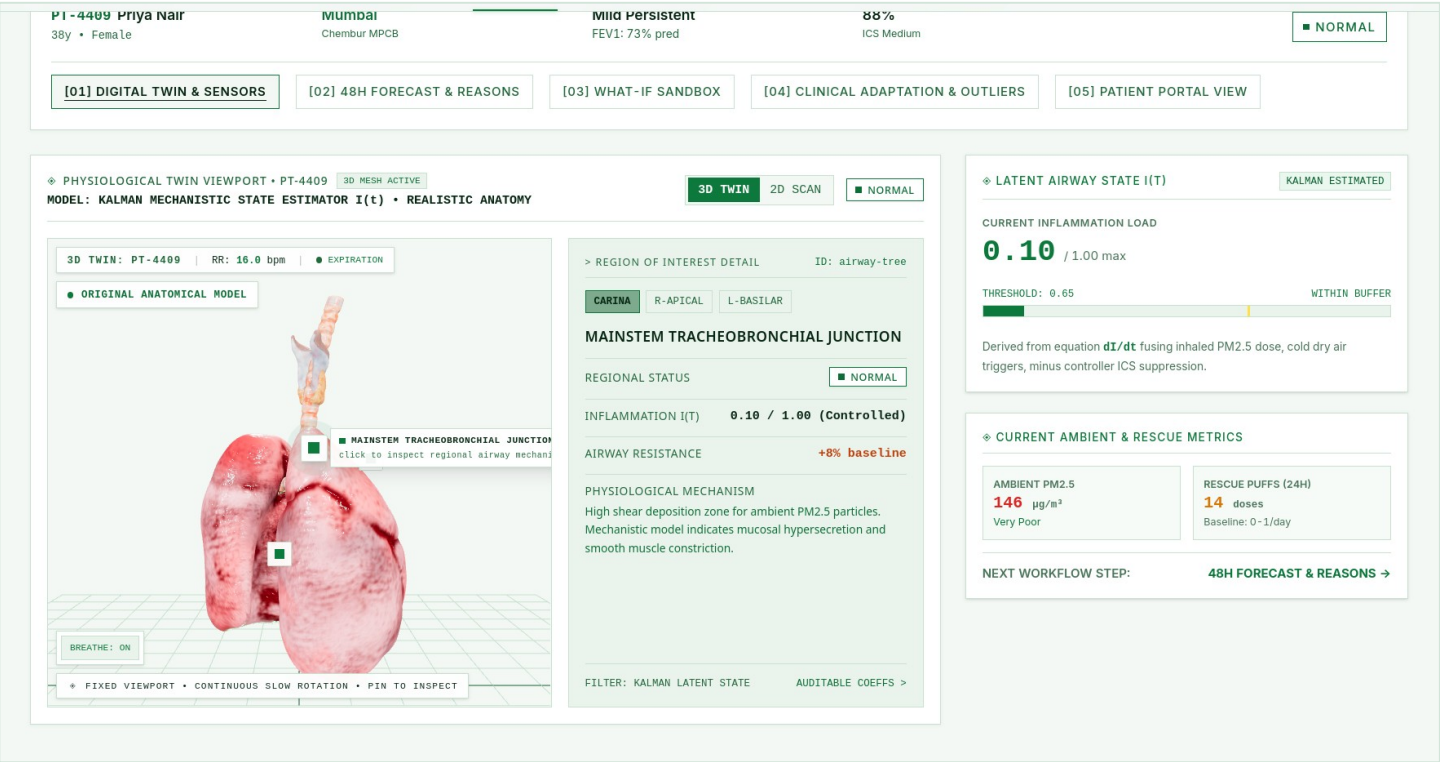


MECHANISTIC CONTEXT A separate airway ODE estimates the latent inflammation state and supports counterfactual re-scoring alongside the learned ensemble.

MECHANISTIC SIMULATION

The twin tracks a modelled state the wearable cannot measure directly

A differential-equation simulator updates latent airway inflammation using exposure, symptoms, systemic triggers and controller effects.



LATENT STATE I(t)

A continuously updated estimate of airway inflammation, bounded in the project model from 0.05 to 1.0.

PROJECT THRESHOLD: 0.65

HOW THE STATE CHANGES

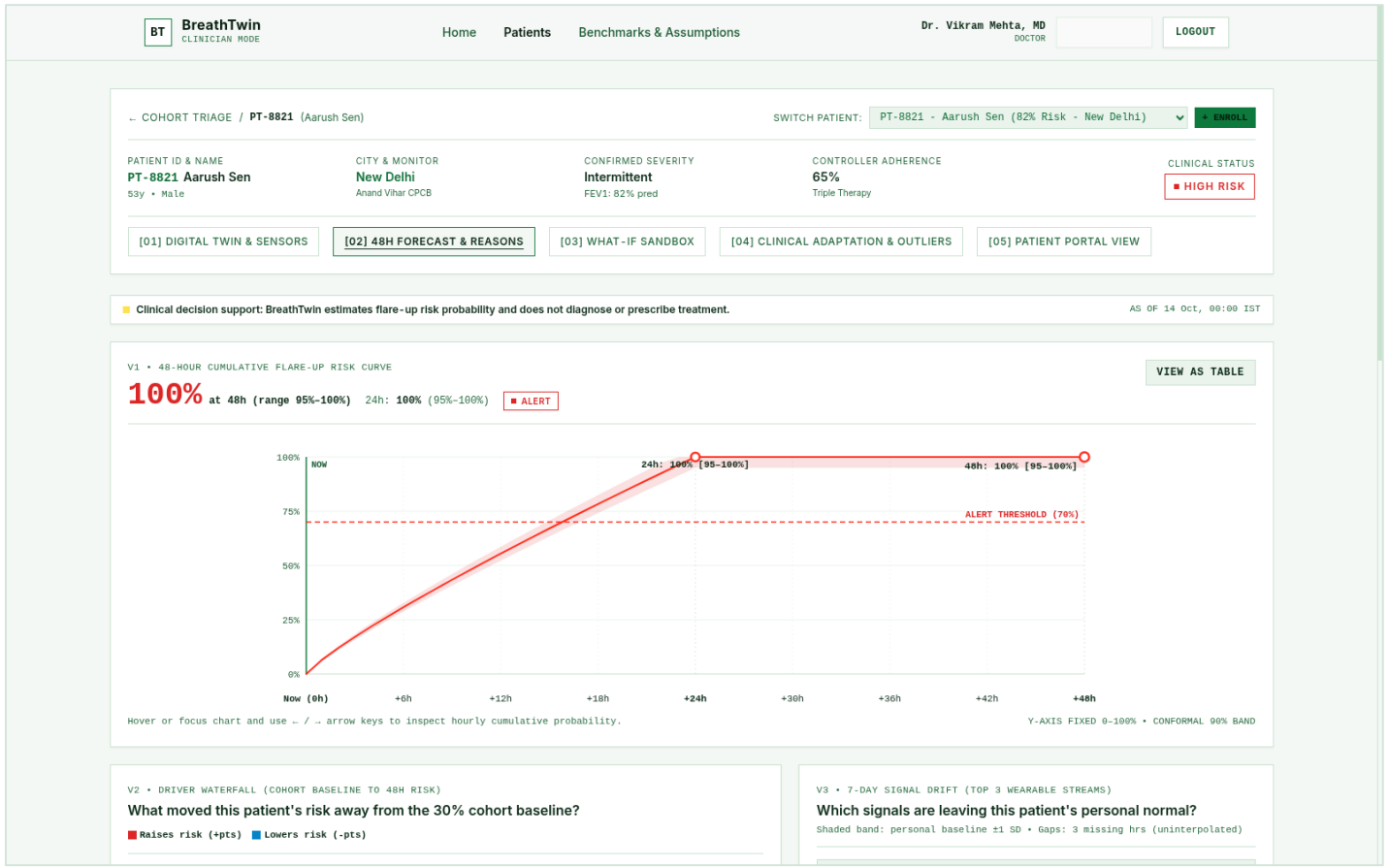
Recovery	tends to lower inflammation
PM2.5 dose	can raise the modelled load
Triggers / illness	add extra perturbations
Controller effect	models anti-inflammatory influence

This is a simulated state, not a direct inflammation measurement or a clinically established universal threshold.

FORECAST AND EXPLANATION

Show the risk trajectory and the signals behind it

BreathTwin presents a forecast as a timeline with uncertainty and local driver attribution, rather than only a single risk number.



01

Two forecast horizons

Separate estimates for 24-hour flare-up risk and 48-hour severe exacerbation risk.

02

Uncertainty is visible

The project uses 90% conformal prediction intervals to show a range around the forecast.

03

Risk drivers are inspectable

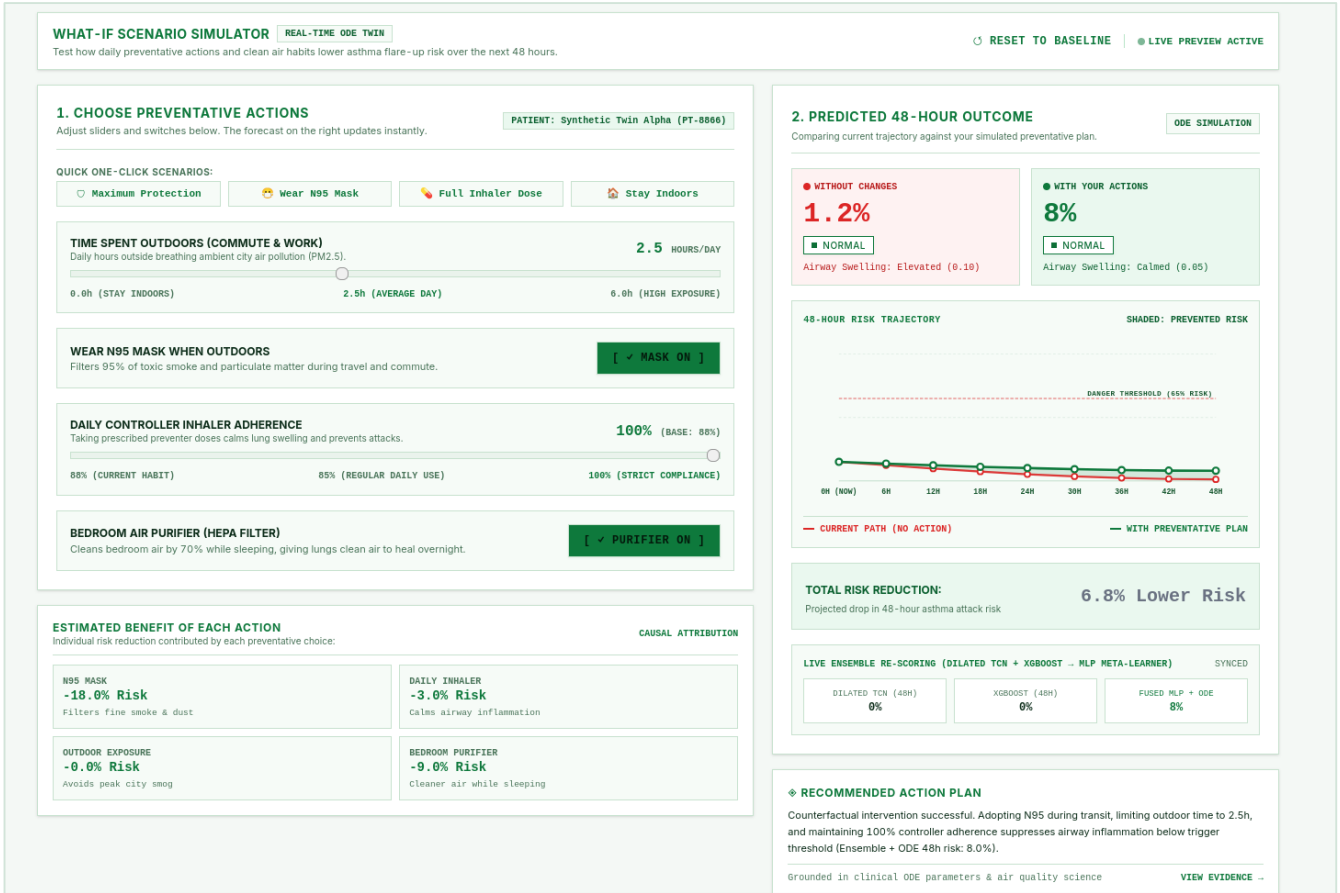
A local SHAP view ranks contributing factors such as rescue use, oxygen changes and particulate exposure.

Conformal coverage must be validated on real-world data.

INTERVENTION SANDBOX

Ask “what if?” before comparing a prevention plan

The sandbox re-runs the model under changed assumptions and displays how the simulated trajectory differs from the current path.



SCENARIOS IN THE PROJECT

Outdoor time	Reduce the hours spent in polluted outdoor air.
N95 mask	Model filtration during commute and outdoor exposure.
HEPA purifier	Model lower indoor particulate exposure.
Controller adherence	Explore adherence assumptions within the care plan.

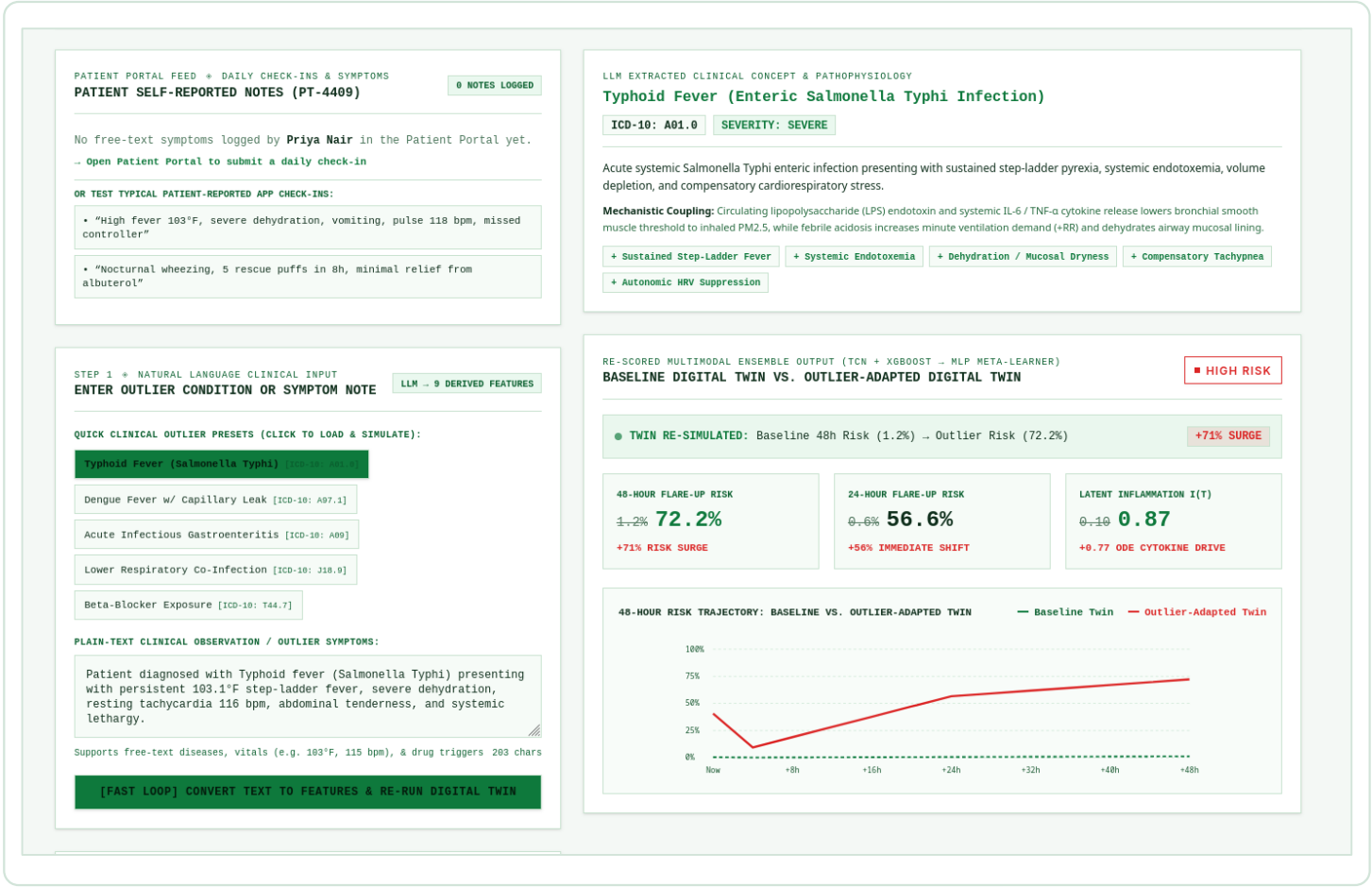
Important boundary

These are modelled counterfactuals, not proof that an intervention will produce the same effect in a real patient. Medication changes remain clinician-led.

CLINICAL NOTE ADAPTATION

Add context when the sensor stream is not the whole story

The prototype translates a clinician's free-text note into structured features, then re-scores the twin with those additional perturbations.



1 Enter a clinical note

For example, a concurrent infection, fever, dehydration or a medication change.

2 Extract structured features

The LLM / ontology pipeline maps text to condition labels, vital-sign changes and ODE modifiers.

3 Re-simulate and review

The twin updates its trajectory; a clinician can inspect the derived features and result.

4 Commit and adapt

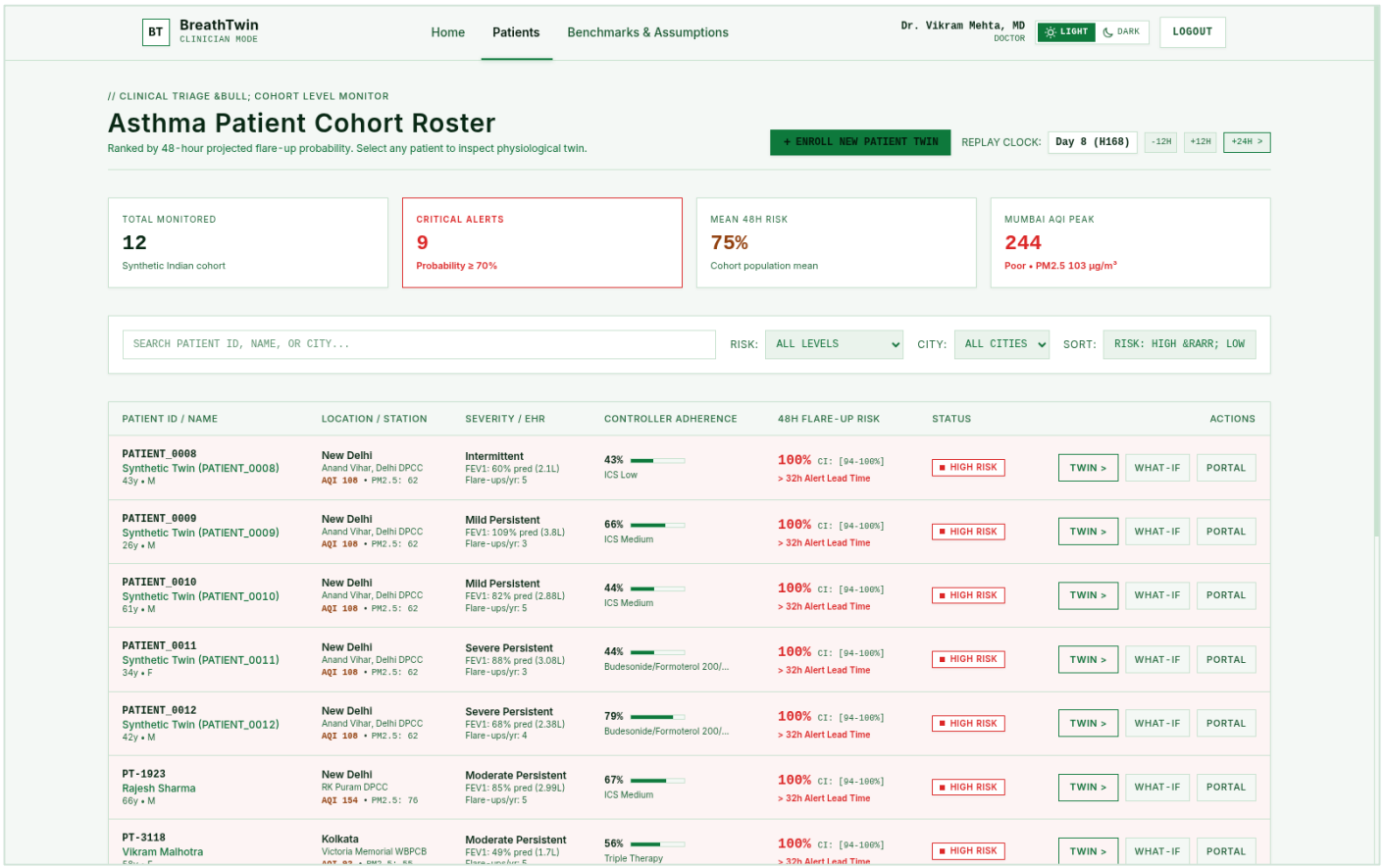
Verified outlier cases enter the replay buffer; EWC regularization aims to reduce forgetting during retraining.

A clinician review and commit step is part of the stated workflow; extracted values should not be treated as verified facts automatically.

DOCTOR WORKFLOW

Start with the cohort, then inspect an individual patient

The clinician interface is designed to help a team prioritize attention, open an individual twin, and review forecasts in context.



Illustrative dashboard values are from the supplied project screenshots and synthetic cohort.

COHORT VIEW

- Rank the patient roster by projected 48-hour risk.
- Filter by city, risk level and available patient context.
- Move from the roster to a 3D twin, forecast, what-if sandbox or note adaptation view.

INDIVIDUAL VIEW

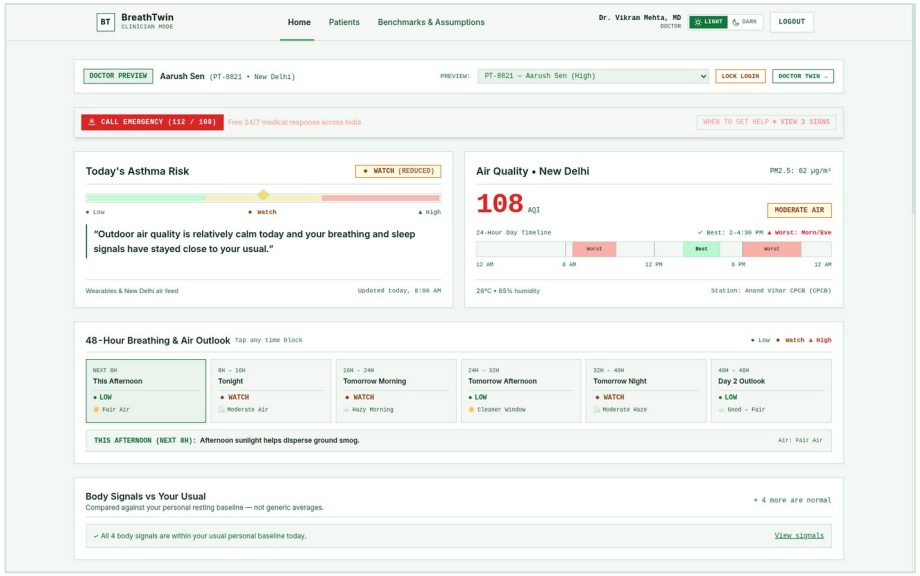
- Compare recent signals with a personal baseline.
- Review the drivers and projected trajectory before discussing a care plan.

PATIENT COMPANION

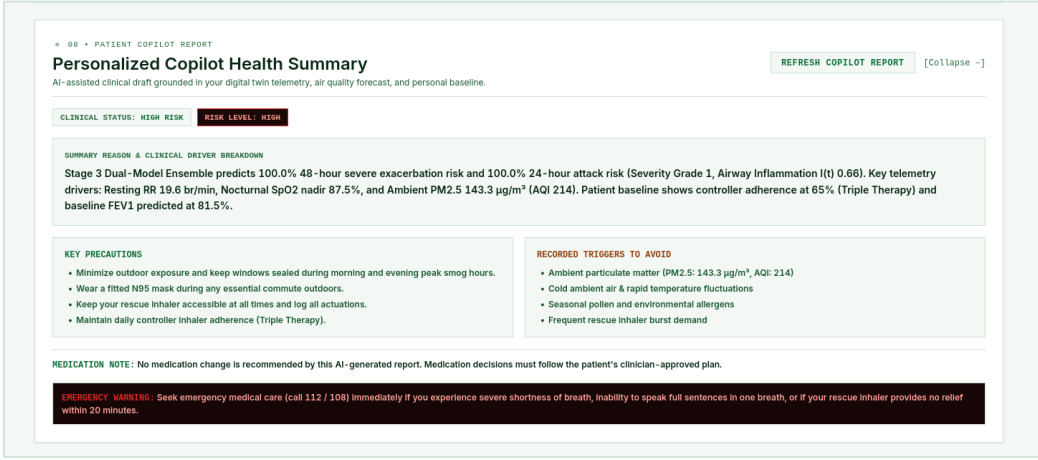
Turn model output into guidance people can understand

The patient portal avoids technical model language and brings risk status, local air outlook, personal signals and daily check-ins into one view.

01 / DAILY STATUS & AIR OUTLOOK



02 / COPILOT HEALTH SUMMARY



Plain-language risk

A simple status and a short explanation of likely drivers.

Daily self-report

Logs symptoms, rescue use and controller adherence.

Safety guardrails

No autonomous prescribing; emergency red flags remain explicit.

REPORTED EVALUATION

Strong separation on the project's synthetic held-out cohort

The repository reports patient-level splits with 800 training, 100 validation and 100 held-out test patients (random seed 42).

0.9399
24-hour ROC-AUC
BreathTwin ensemble

0.9428
48-hour ROC-AUC
BreathTwin ensemble

0.8435
Severity weighted F1
4 severity grades

< 65 ms
Counterfactual re-score
Reported simulation latency

ROC-AUC COMPARISON

Model	24 h ROC-AUC	48 h ROC-AUC	Mean warning lead time
BreathTwin (TCN + XGBoost + MLP + ODE)	**0.9399**	**0.9428**	**42.4 ± 4.1 h**
LSTM baseline	0.7850	0.7790	22.1 ± 5.8 h
XGBoost-only baseline	0.7410	0.7320	18.6 ± 6.2 h

Interpretation limit These are project-reported results on synthetic data. They do not establish performance, clinical utility or safety on real patients.

Source: BreathTwin README, §7.6 (held-out test metrics). Values rounded only where shown in stat cards.

SAFETY AND SCOPE

Built as decision support, with important limits made explicit

The prototype includes guardrails, but its forecasts and simulated intervention effects require additional validation before clinical use.

DESIGN SAFEGUARDS

Synthetic patient records

The repository states that patient records are synthetic; do not interpret the screenshots as real patient cases.

Clinician stays in control

BreathTwin is framed as decision support, not autonomous diagnosis or prescribing.

Role separation and logs

Doctor and patient access are separated through role-based access and session controls; events are logged.

Non-prescriptive copilot

Patient-facing output reinforces the clinician-approved plan and includes emergency red flags.

WHAT IS NOT YET PROVEN

Real-world prediction accuracy

Synthetic test metrics do not show how the model performs in clinical populations.

Calibrated uncertainty in practice

Prediction intervals need coverage and calibration checks on representative real-world data.

Causal benefit of actions

A simulated risk decrease is not proof an intervention will reduce a person's risk by the same amount.

Clinical workflow readiness

Prospective evaluation, governance, privacy review and clinician assessment are still required.

PROJECT HANDOFF

An open-source prototype built by an inter-college team

BreathTwin brings the model pipeline, physiological simulator, application interfaces and reproducible local setup into one repository.

TECHNICAL STACK

Frontend	Next.js 16 · React 19 · TypeScript · Tailwind CSS
3D experience	Three.js · WebGL · GLSL shaders
API and services	Python · FastAPI · Pydantic · Uvicorn
Models	PyTorch · Dilated TCN · XGBoost · Stacking MLP
Simulation and data	Mechanistic ODE · PostgreSQL 16 · SQLAlchemy
Delivery and tests	Docker Compose · Pytest · run.sh

TEAM & CONTRIBUTIONS

Aarush Bhattacharjee

LNMIIT · Team lead, system architecture, backend / full-stack, mechanistic simulator and copilot integration.

Rakshit Gupta

NSUT · Machine learning engineer, TCN, XGBoost, stacking model, preprocessing and benchmark validation.

MIT LICENSE

SOURCE CODE github.com/Frisbee148/DigitalTwin

Connect signals → estimate state → explain risk → compare scenarios → keep clinicians in control